Class #14 - Least-Squares Problems

1. Motivation: Suppose $Ax = b$ is inconsistent. Find x such that Ax best approximates b.
   That is, the smaller the distance between b and Ax (_____), the better the approximation.

2. The general least-squares problem: Find x that makes
   $||b - Ax||$ as small as possible.

3. Def: If $A \in \mathbf{R^{m \times n}}$ and $b \in \mathbf{R^n}$, a *least-squares solution* of $Ax = b$ is an $\hat{x} \in \mathbf{R^n}$ such that

$$||b - A\hat{x}|| \leq ||b - Ax||$$

   for all $x \in \mathbf{R^n}$.

4. Observation: For any $x \in \mathbf{R^n}$, $Ax \in$ _____.
   The closest point to b in Col$A$ = _____.
   Note: The equation $Ax = \hat{b}$ is _____. Hence, the least-squares solution _____.

5. Solution of the General Least-Squares Problem:
   A vector $\hat{x}$ is a least-squares solution of $Ax = b$ if and only if A_____ = _____.
   Suppose $\hat{x}$ satisfies $A\hat{x} = \hat{b}$.
   Note: $(b - \hat{b}) \in$ _____ = (Row(A^T))$^\perp$ = _____(A^T).
   Hence, $A^T(b - A\hat{x}) =$ _____, that is,

$$A^T A\hat{x} = A^T b.$$

   (The above equation is commonly referred to as the *normal equations for* $\hat{x}$.)
   The solution set to the normal equations for $\hat{x}$ is _____.

6. Example: Find a least-squares solution of the inconsistent system $Ax = b$ for

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} -3 \\ -1 \\ 0 \\ 2 \\ 5 \\ 1 \end{bmatrix}.$$

$$A^T A = \begin{bmatrix} 6 & 2 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 2 & 0 & 0 & 2 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 4 \\ -4 \\ 2 \\ 6 \end{bmatrix}.$$

The augmented matrix for the normal equations for $\hat{x}$ is row equivalent to

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & -1 & -5 \\ 0 & 0 & 1 & -1 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

The general least-squares solution of $Ax = b$ has the form:

7. Question: When is there only one least-squares solution?

8. Theorem: The matrix $A^T A$ is invertible if and only if the columns of A are _____. In this case, the equation $Ax = b$ has _____least-squares solution $\hat{x}$, and it is given by _____.

   Proof:
   Claim #1: Nul A = Nul $A^T A$
   (Proof of Claim #1):

...

Claim #2: If A is an $m \times n$ matrix such that A^TA is invertible, then the columns of A are linearly independent.
(Proof of Claim #2):

Claim #3: If A is an $m \times n$ matrix whose columns are linearly independent, then A^TA is invertible.
(Proof of Claim #3):

Hence, the 1st statement of the Theorem is true, and the 2nd statement follows.

9. Theorem: Given an $m \times n$ matrix with linearly independent columns (then $A =$ _____), for each $b \in \mathbf{R^n}$, the equation $Ax = b$ has a _____least-squares solution, given by

$$\hat{x} = R^{-1}Q^Tb.$$

Proof:

Note: Since R is upper trianguar, it is easier to calculate _____.

10. Def: When a least-squares solution $\hat{x}$ is used to produce $A\hat{x}$ as an approximation to b, the distance from b to $A\hat{x}$ (_____) is called the *least-squares error* of this approximation.